

CHAPTER 8

the Black–Scholes formulae and the ‘greeks’



The aim of this Chapter...

...is to show how the basic Black–Scholes formulae are derived from the Black–Scholes equation, and to introduce more sophisticated hedging strategies. The chapter contains lots of useful formulae which are also summarized at the end.

In this Chapter...

- the derivation of the Black–Scholes formulae for calls, puts and simple digitals
- the meaning and importance of the ‘greeks,’ delta, gamma, theta, vega and rho
- the difference between differentiation with respect to variables and to parameters
- formulae for the greeks for calls, puts and simple digitals

8.1 INTRODUCTION

The Black–Scholes equation has simple solutions for calls, puts and some other contracts. In this chapter I’m going to walk you through the derivation of these formulæ step by step. This is one of the few places in the book where I do derive formulæ. The reason that I don’t often derive formulæ is that the majority of contracts do not have explicit solutions for their theoretical value. Instead much of my emphasis will be placed on finding numerical solutions of the Black–Scholes equation.

We’ve seen how the quantity ‘delta,’ the first derivative of the option value with respect to the underlying, occurs as an important quantity in the derivation of the Black–Scholes equation. In this chapter I describe the importance of other derivatives of the option price, with respect to the variables (the underlying asset and time) and with respect to some of the parameters. These derivatives are important in the hedging of an option position, playing key roles in risk management. It can be argued that it is more important to get the hedging correct than to be precise in the pricing of a contract. The reason for this is that if you are accurate in your hedging you will have reduced or eliminated future uncertainty. This leaves you with a profit (or loss) that is set the moment that you buy or sell the contract. But if your hedging is inaccurate, then it doesn’t matter, within reason, what you sold the contract for initially, future uncertainty could easily dominate any initial profit. Of course, life is not so simple, in reality we are exposed to model error, which can make a mockery of anything we do. However, this illustrates the importance of good hedging, and that’s where the ‘greeks’ come in.



Time Out...

Close your eyes until I tell you to open them

Unless you are doing a highly mathsy course, you won’t need to know all the manipulations that follow.

8.2 DERIVATION OF THE FORMULÆ FOR CALLS, PUTS AND SIMPLE DIGITALS

The Black–Scholes equation is

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0. \quad (8.1)$$

This equation must be solved with final condition depending on the payoff: each contract will have a different functional form prescribed at expiry $t = T$, depending on whether it is a call, a put or something more fancy. This is the final condition that must be imposed

to make the solution unique. We'll worry about final conditions later, for the moment concentrate on manipulating (8.1) into something we can easily solve.

The first step in the manipulation is to change from present value to future value terms. Recalling that the payoff is received at time T but that we are valuing the option at time t this suggests that we write

$$V(S, t) = e^{-r(T-t)}U(S, t).$$

This takes our differential equation to

$$\frac{\partial U}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + rS \frac{\partial U}{\partial S} = 0.$$

The second step is really trivial. Because we are solving a backward equation, discussed in Chapter 7, we'll write

$$\tau = T - t.$$

This now takes our equation to

$$\frac{\partial U}{\partial \tau} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + rS \frac{\partial U}{\partial S}.$$

When we first started modeling equity prices we used intuition about the asset price *return* to build up the stochastic differential equation model. Let's go back to examine the return and write

$$\xi = \log S.$$

With this as the new variable, we find that

$$\frac{\partial}{\partial S} = e^{-\xi} \frac{\partial}{\partial \xi} \quad \text{and} \quad \frac{\partial^2}{\partial S^2} = e^{-2\xi} \frac{\partial^2}{\partial \xi^2} - e^{-2\xi} \frac{\partial}{\partial \xi}.$$

Now the Black–Scholes equation becomes

$$\frac{\partial U}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 U}{\partial \xi^2} + \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial U}{\partial \xi}.$$

What has this done for us? It has taken the problem defined for $0 \leq S < \infty$ to one defined for $-\infty < \xi < \infty$. But more importantly, the coefficients in the equation are now all constant, independent of the underlying. This is a big step forward, made possible by the lognormality of the underlying asset. We are nearly there.

The last step is simple, but the motivation is not so obvious. Write

$$x = \xi + \left(r - \frac{1}{2}\sigma^2\right) \tau,$$

and $U = W(x, \tau)$. This is just a 'translation' of the coordinate system. It's a bit like using the forward price of the asset instead of the spot price as a variable. After this change of variables the Black–Scholes becomes the simpler

$$\frac{\partial W}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 W}{\partial x^2}. \tag{8.2}$$

To summarize,

$$\begin{aligned} V(S, t) &= e^{-r(T-t)} U(S, t) = e^{-r\tau} U(S, T - \tau) = e^{-r\tau} U(e^{\xi}, T - \tau) \\ &= e^{-r\tau} U\left(e^{x - \left(r - \frac{1}{2}\sigma^2\right)\tau}, T - \tau\right) = e^{-r\tau} W(x, \tau). \end{aligned}$$

To those of you who already know the Black–Scholes formulæ for calls and puts the variable x will ring a bell:

$$x = \xi + \left(r - \frac{1}{2}\sigma^2\right)\tau = \log S + \left(r - \frac{1}{2}\sigma^2\right)(T - t).$$

Having turned the original Black–Scholes equation into something much simpler, let's take a break for a moment while I explain where we are headed.

I'm going to derive an expression for the value of any option whose payoff is a known function of the asset price at expiry. This includes calls, puts and digitals. This expression will be in the form of an integral. For special cases, I'll show how to rewrite this integral in terms of the cumulative distribution function for the Normal distribution. This is particularly useful since the function can be found on spreadsheets, calculators and in the backs of books. But there are two steps before I can write down this integral.

The first step is to find a special solution of (8.2), called the fundamental solution. This solution has useful properties. The second step is to use the linearity of the equation and the useful properties of the special solution to find the *general solution* of the equation. Here we go.

I'm going to look for a special solution of (8.2) of the following form

$$W(x, \tau) = \tau^{\alpha} f\left(\frac{(x - x')}{\tau^{\beta}}\right), \quad (8.3)$$

where x' is an arbitrary constant. And I'll call this special solution $W_f(x, \tau; x')$. Note that the unknown function depends on only *one* variable $(x - x')/\tau^{\beta}$. As well as finding the function f we must find the constant parameters α and β . We can expect that if this approach works, the equation for f will be an ordinary differential equation since the function only has one variable. This reduction of dimension is an example of a similarity reduction, discussed in Chapter 7.

Substituting expression (8.3) into (8.2) we get

$$\tau^{\alpha-1} \left(\alpha f - \beta \eta \frac{df}{d\eta} \right) = \frac{1}{2} \sigma^2 \tau^{\alpha-2\beta} \frac{d^2 f}{d\eta^2}, \quad (8.4)$$

where

$$\eta = \frac{x - x'}{\tau^{\beta}}.$$

Examining the dependence of the two terms in (8.4) on both τ and η we see that we can only have a solution if

$$\alpha - 1 = \alpha - 2\beta \quad \text{i.e.} \quad \beta = \frac{1}{2}.$$

I want to ensure that my 'special solution' has the property that its integral over all ξ is independent of τ , for reasons that will become apparent. To ensure this, I require

$$\int_{-\infty}^{\infty} \tau^{\alpha} f((x - x')/\tau^{\beta}) dx$$

to be constant. I can write this as

$$\int_{-\infty}^{\infty} \tau^{\alpha+\beta} f(\eta) d\eta$$

and so I need

$$\alpha = -\beta = -\frac{1}{2}.$$

The function f now satisfies

$$-f - \eta \frac{df}{d\eta} = \sigma^2 \frac{d^2 f}{d\eta^2}.$$

This can be written

$$\sigma^2 \frac{d^2 f}{d\eta^2} + \frac{d(\eta f)}{d\eta} = 0,$$

which can be integrated once to give

$$\sigma^2 \frac{df}{d\eta} + \eta f = a,$$

where a is a constant. For my special solution I'm going to choose $a = 0$. This equation can be integrated again to give

$$f(\eta) = b e^{-\frac{\eta^2}{2\sigma^2}}.$$

I will choose the constant b such that the integral of f from minus infinity to plus infinity is one:

$$f(\eta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\eta^2}{2\sigma^2}}.$$

This is the special solution I have been seeking:¹

$$W(x, \tau) = \frac{1}{\sqrt{2\pi\tau}\sigma} e^{-\frac{(x-x')^2}{2\sigma^2\tau}}.$$

Now I will explain why it is useful in our quest for the Black–Scholes formulæ.

In Figure 8.1 is plotted W as a function of x' for several values of τ . Observe how the function rises in the middle but decays at the sides. As $\tau \rightarrow 0$ this becomes more pronounced. The 'middle' is the point $x' = x$. At this point the function grows unboundedly

¹ It is just the probability density function for a Normal random variable with mean zero and standard deviation σ .

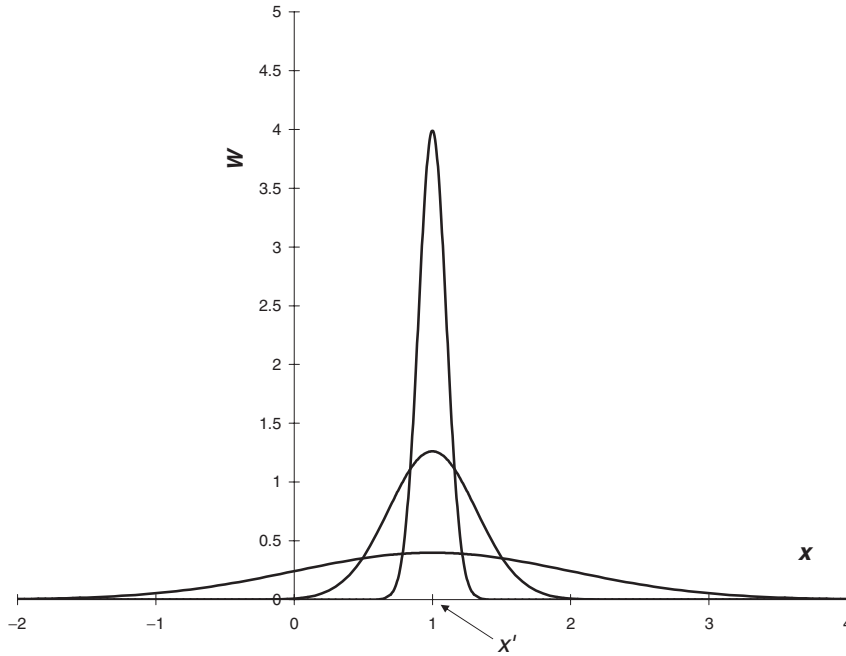


Figure 8.1 The fundamental solution.

and away from this point the function decays to zero as $\tau \rightarrow 0$. Although the function is increasingly confined to a narrower and narrower region its area remains fixed at one. These properties of decay away from one point, unbounded growth at that point and constant area, result in a **Dirac delta function** $\delta(x' - x)$ as $\tau \rightarrow 0$. The delta function has one important property, namely

$$\int \delta(x' - x) g(x') dx' = g(x)$$

where the integration is from any point below x to any point above x . Thus the delta function ‘picks out’ the value of g at the point where the delta function is singular, i.e. at $x' = x$. In the limit as $\tau \rightarrow 0$ the function W becomes a delta function at $x = x'$. This means that

$$\lim_{\tau \rightarrow 0} \frac{1}{\sigma \sqrt{2\pi\tau}} \int_{-\infty}^{\infty} e^{-\frac{(x'-x)^2}{2\sigma^2\tau}} g(x') dx' = g(x).$$

This property of the special solution, together with the linearity of the Black–Scholes equation, are all that are needed to find some explicit solutions.

Now is the time to consider the payoff. Let’s call it

$$\text{Payoff}(S).$$

This is the value of the option at time $t = T$. It is the final condition for the function V , satisfying the Black–Scholes equation:

$$V(S, T) = \text{Payoff}(S).$$

In our new variables, this final condition is

$$W(x, 0) = \text{Payoff}(e^x). \quad (8.5)$$

I claim that the solution of this for $\tau > 0$ is

$$W(x, \tau) = \int_{-\infty}^{\infty} W_f(x, \tau; x') \text{Payoff}(e^{x'}) dx'. \quad (8.6)$$

To show this, I just have to demonstrate that the expression satisfies Equation (8.2) and the final condition (8.5). Both of these are straightforward. The integration with respect to x' is similar to a summation, and since each individual component satisfies the equation so does the sum/integral. Alternatively, differentiate (8.6) under the integral sign to see that it satisfies the partial differential equation. That it satisfies condition (8.5) follows from the special properties of the fundamental solution W_f .

Retracing our steps to write our solution in terms of the original variables, we get

$$V(S, t) = \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}} \int_0^{\infty} e^{-\left(\log(S/S') + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 / 2\sigma^2(T-t)} \text{Payoff}(S') \frac{dS'}{S'}, \quad (8.7)$$

where I have written $x' = \log S'$.

This is the exact solution for the option value in terms of the arbitrary payoff function. In the next sections I will manipulate this expression for special payoff functions.

8.2.1 Formula for a call

The call option has the payoff function

$$\text{Payoff}(S) = \max(S - E, 0).$$

Expression (8.7) can then be written as

$$\frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}} \int_E^{\infty} e^{-\left(\log(S/S') + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 / 2\sigma^2(T-t)} (S' - E) \frac{dS'}{S'}.$$

Return to the variable $x' = \log S'$, to write this as

$$\begin{aligned} & \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}} \int_{\log E}^{\infty} e^{-\left(-x' + \log S + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 / 2\sigma^2(T-t)} (e^{x'} - E) dx' \\ &= \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}} \int_{\log E}^{\infty} e^{-\left(-x' + \log S + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 / 2\sigma^2(T-t)} e^{x'} dx' \\ & \quad - E \frac{e^{-r(T-t)}}{\sigma\sqrt{2\pi(T-t)}} \int_{\log E}^{\infty} e^{-\left(-x' + \log S + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 / 2\sigma^2(T-t)} dx'. \end{aligned}$$

Both integrals in this expression can be written in the form

$$\int_d^{\infty} e^{-\frac{1}{2}x'^2} dx'$$

for some d (the second is just about in this form already, and the first just needs a completion of the square).



Time Out...

You can open your eyes now

From now on I just quote formulæ without giving derivations.

Apart from a couple of minor differences, this integral is just like the cumulative distribution function for the standardized Normal distribution² defined by

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}\phi^2} d\phi.$$

This function, plotted in Figure 8.2, is the probability that a Normally distributed variable is less than x .

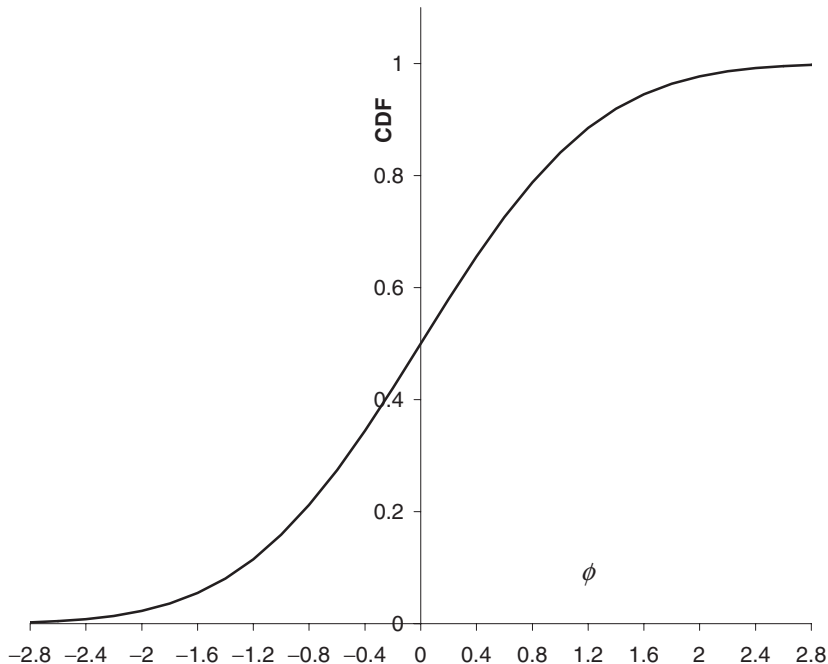


Figure 8.2 The cumulative distribution function for a standardized Normal random variable, $N(x)$.

² I.e. having zero mean and unit standard deviation.

Thus the option price can be written as two separate terms involving the cumulative distribution function for a Normal distribution:

$$\text{Call option value} = SN(d_1) - Ee^{-r(T-t)}N(d_2)$$

where

$$d_1 = \frac{\log(S/E) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

and

$$d_2 = \frac{\log(S/E) + (r - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}.$$

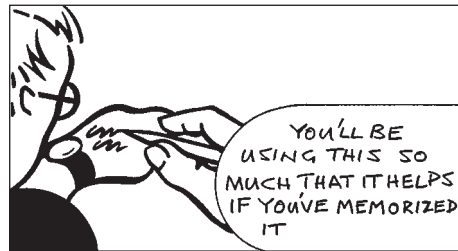
When there is continuous dividend yield on the underlying, or it is a currency, then



Excel representation of call value

Call option value

$$\begin{aligned} & Se^{-D(T-t)}N(d_1) - Ee^{-r(T-t)}N(d_2) \\ d_1 &= \frac{\log(S/E) + (r - D + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \\ d_2 &= \frac{\log(S/E) + (r - D - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \\ &= d_1 - \sigma\sqrt{T-t} \end{aligned}$$



Time Out...

In Excel

In Excel the cumulative distribution function for the standardized Normal distribution is NORMSDIST(). The natural logarithm is LN().



The option value is shown in Figure 8.3 as a function of the underlying asset at a fixed time to expiry. In Figure 8.4 the value of the at-the-money option is shown as a function of time, expiry is $t = 1$. In Figure 8.5 is the call value as a function of both the underlying and time.

When the asset is 'at-the-money forward,' i.e. $S = Ee^{-(r-D)(T-t)}$, then there is a simple approximation for the call value (Brenner & Subrahmanyam, 1994):

$$\text{Call} \approx 0.4 Se^{-D(T-t)}\sigma\sqrt{T-t}.$$

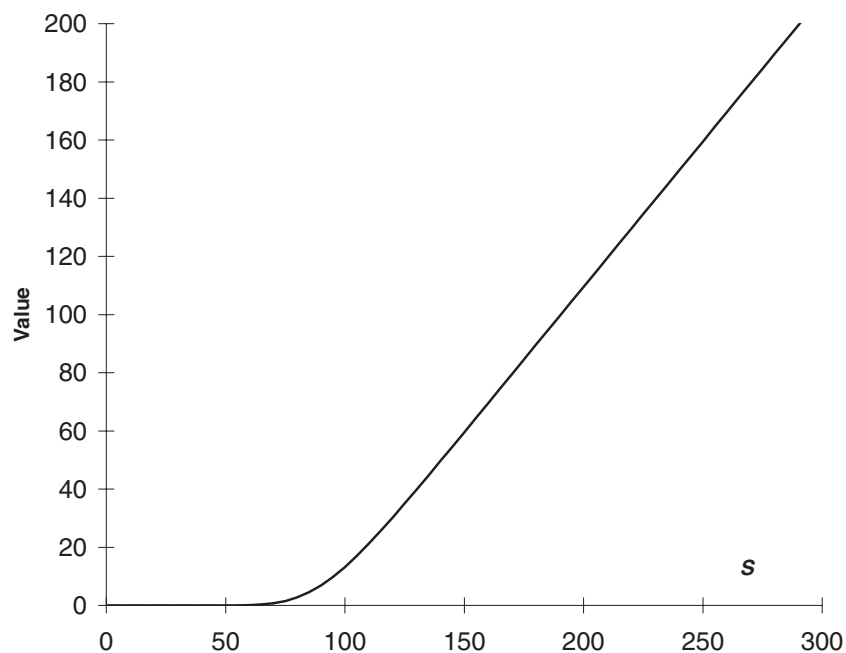


Figure 8.3 The value of a call option as a function of the underlying asset price at a fixed time to expiry.

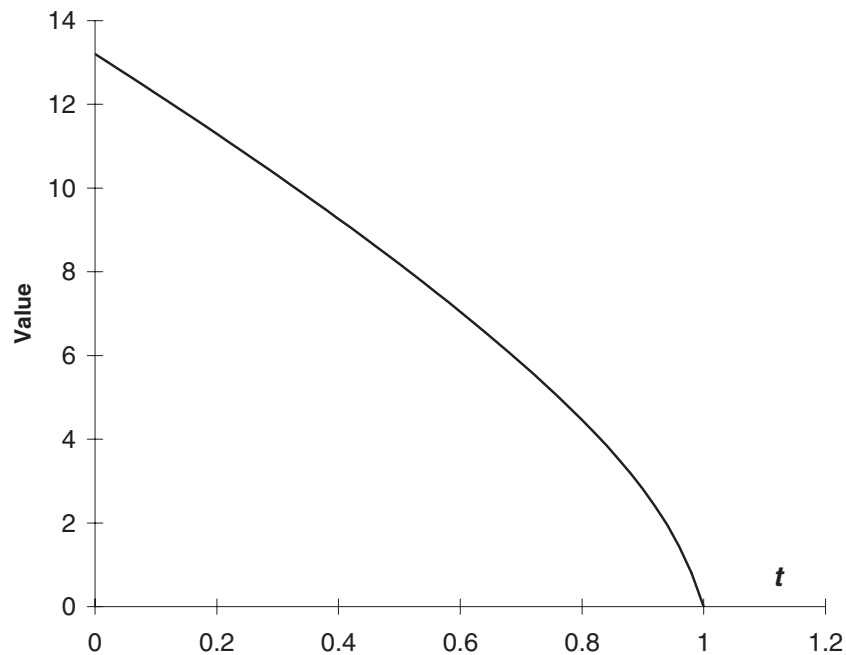


Figure 8.4 The value of an at-the-money call option as a function of time.

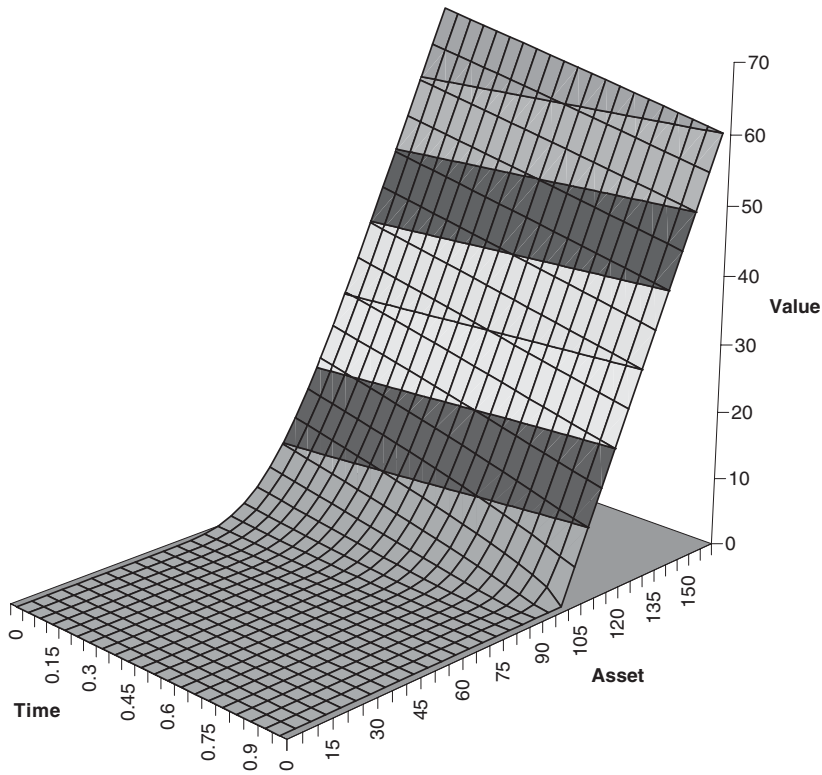


Figure 8.5 The value of a call option as a function of asset and time.

8.2.2 Formula for a put

The put option has payoff

$$\text{Payoff}(S) = \max(E - S, 0).$$

The value of a put option can be found in the same way as above, or using put-call parity

$$\text{Put option value} = -SN(-d_1) + Ee^{-r(T-t)}N(-d_2),$$

with the same d_1 and d_2 .

When there is continuous dividend yield on the underlying, or it is a currency, then

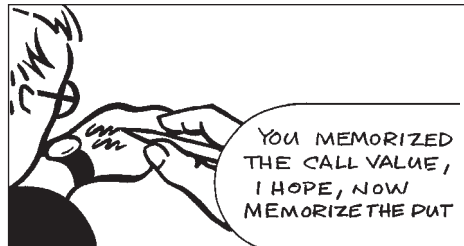
Put option value

$$-Se^{-D(T-t)}N(-d_1) + Ee^{-r(T-t)}N(-d_2)$$

The option value is shown in Figure 8.6 against the underlying asset and in Figure 8.7 against time. In Figure 8.8 is the option value as a function of both the underlying asset and time.



Excel representation of put value



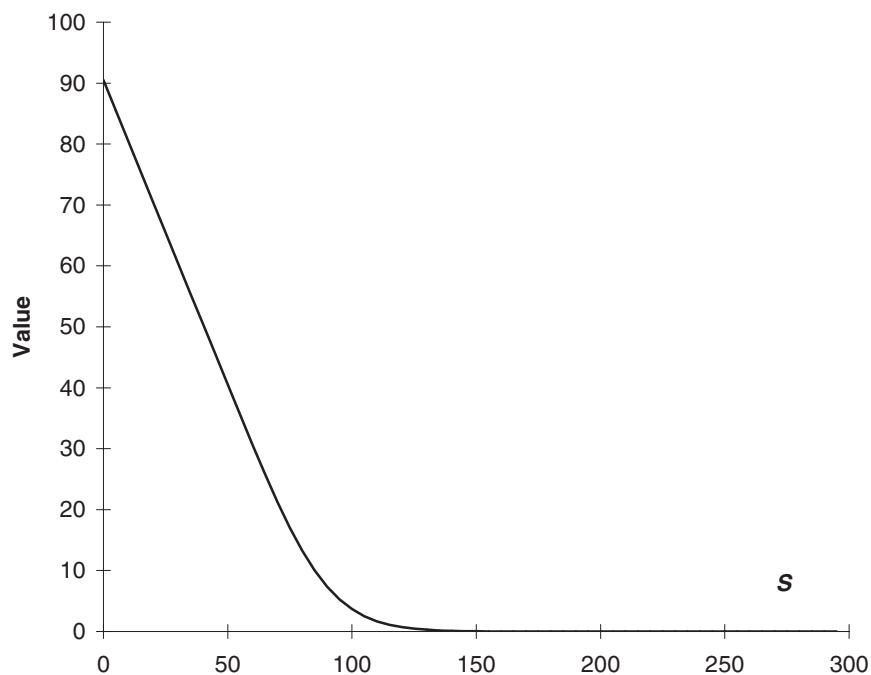


Figure 8.6 The value of a put option as a function of the underlying asset at a fixed time to expiry.

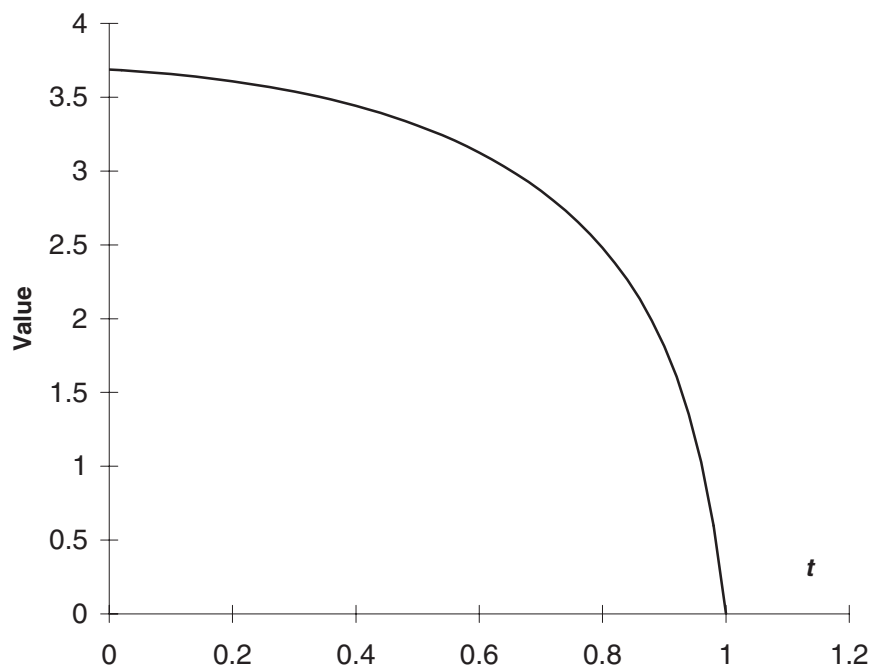


Figure 8.7 The value of an at-the-money put option as a function of time.

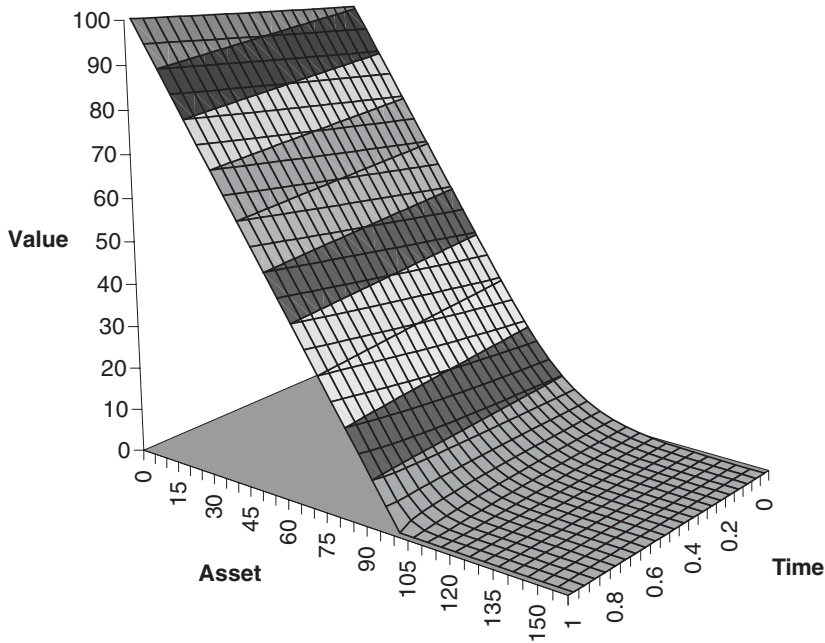


Figure 8.8 The value of a put option as a function of asset and time.

When the asset is at-the-money forward the simple approximation for the put value (Brenner & Subrahmanyam, 1994) is

$$\text{Put} \approx 0.4 S e^{-D(T-t)} \sigma \sqrt{T-t}.$$

8.2.3 Formula for a binary call

The binary call has payoff

$$\text{Payoff}(S) = \mathcal{H}(S - E),$$

where $\mathcal{H}$ is the Heaviside function taking the value one when its argument is positive and zero otherwise.

Incorporating a dividend yield, we can write the option value as

$$\frac{e^{-r(T-t)}}{\sigma \sqrt{2\pi(T-t)}} \int_{\log E}^{\infty} e^{-\left(x' - \log S - \left(r - D - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 / 2\sigma^2(T-t)} dx'.$$

This term is just like the second term in the call option equation and so

Binary call option value

$$e^{-r(T-t)} N(d_2)$$

The option value is shown in Figure 8.9.



Excel
representation of
binary values

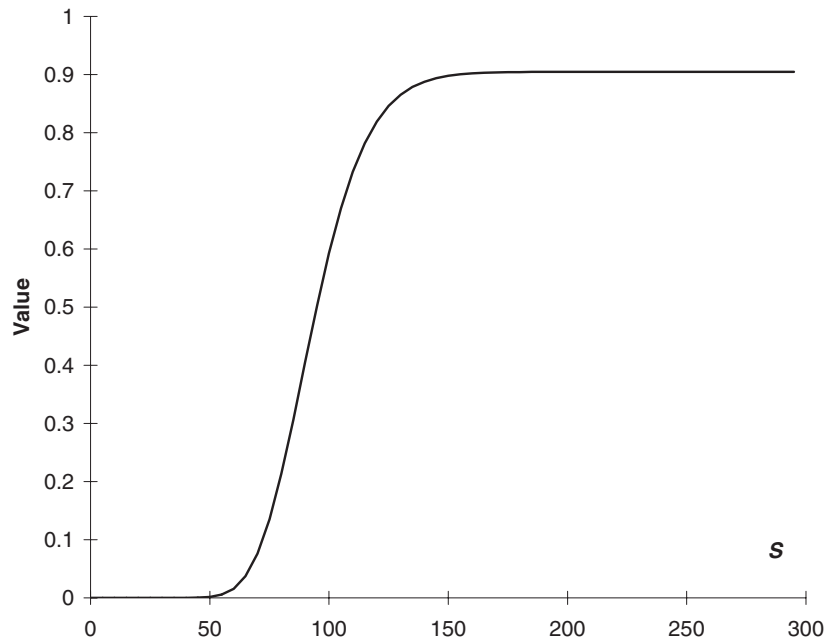


Figure 8.9 The value of a binary call option.

8.2.4 Formula for a binary put

The binary put has a payoff of one if $S < E$ at expiry. It has a value of

Binary put option value

$$e^{-r(T-t)}(1 - N(d_2))$$

since a binary call and a binary put must add up to the present value of \$1 received at time T . The option value is shown in Figure 8.10.



8.3 DELTA

The **delta** of an option or a portfolio of options is the sensitivity of the option or portfolio to the underlying. It is the rate of change of value with respect to the asset:

$$\Delta = \frac{\partial V}{\partial S}$$

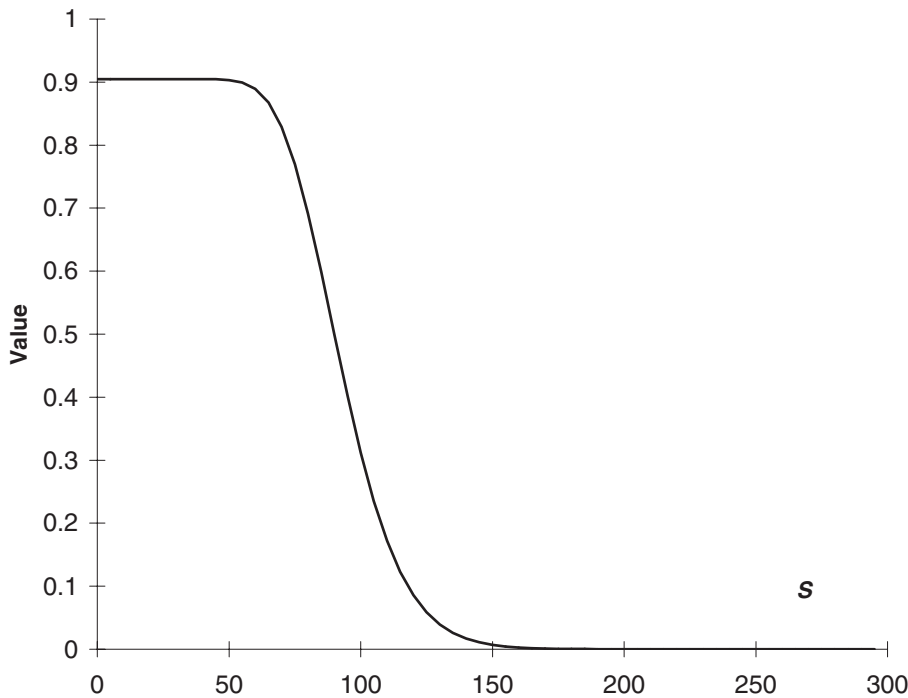


Figure 8.10 The value of a binary put option.

Here V can be the value of a single contract or of a whole portfolio of contracts. The delta of a portfolio of options is just the sum of the deltas of all the individual positions.

The theoretical device of delta hedging, introduced in Chapters 3 and 6, for eliminating risk is far more than that, it is a very important practical technique.

Roughly speaking, the financial world is divided up into speculators and hedgers. The speculators take a view on the direction of some quantity such as the asset price (or more abstract quantities such as volatility) and implement a strategy to take advantage of their view. Such people may not hedge at all.

Then there are the hedgers. There are two kinds of hedger: the ones who hold a position already and want to eliminate some very specific risk (usually using options) and the ones selling (or buying) the options because they believe they have a better price and can make money by hedging away *all* risk. It is the latter type of hedger that is delta hedging. They can only guarantee to make a profit by selling a contract for a high value if they can eliminate all of the risk due to the random fluctuation in the underlying.

Delta hedging means holding one of the option and short a quantity Δ of the underlying. Delta can be expressed as a function of S and t , I'll give some formulae later in this section. This function varies as S and t vary. This means that the number of assets held must be continuously changed to maintain a **delta-neutral** position, this procedure is called **dynamic hedging**. Changing the number of assets held requires the continual purchase and/or sale of the stock. This is called **rehedging** or **rebalancing** the portfolio.

This delta hedging may take place very frequently in highly liquid markets where it is relatively costless to buy and sell. Thus the Black–Scholes assumption of continuous hedging may be quite accurate. In less liquid markets, you lose a lot on bid-offer spread and will therefore hedge less frequently. Moreover, you may not even be able to buy or sell in the quantities you want. Even in the absence of costs, you cannot be sure that your model for the underlying is accurate. There will certainly be some risk associated with the model. These issues make delta hedging less than perfect and in practice the risk in the underlying cannot be hedged away perfectly.

Some contracts have a delta that becomes very large at special times or asset values. The size of the delta makes delta hedging impossible; what can you do if you find yourself with a theoretical delta requiring you to buy more stock than exists? In such a situation the basic foundation of the Black–Scholes world has collapsed and you would be right to question the validity of any pricing formula. This happens at expiry close to the strike for binary options. Although I've given a formula for their price above and a formula for their delta below, I'd be careful using them if I were you.

Here are some formulæ for the deltas of common contracts (all formulæ assume that the underlying pays dividends or is a currency):



Excel
representation of
delta values

Deltas of common contracts

Call	$e^{-D(T-t)}N(d_1)$
Put	$e^{-D(T-t)}(N(d_1) - 1)$
Binary call	$\frac{e^{-r(T-t)}N'(d_2)}{\sigma S\sqrt{T-t}}$
Binary put	$-\frac{e^{-r(T-t)}N'(d_2)}{\sigma S\sqrt{T-t}}$
	$N'(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$

Examples of these functions are plotted in Figure 8.11, with some scaling of the binaries.



8.4 GAMMA

The **gamma**, Γ , of an option or a portfolio of options is the second derivative of the position with respect to the underlying:

$$\Gamma = \frac{\partial^2 V}{\partial S^2}$$

Since gamma is the sensitivity of the delta to the underlying it is a measure of how much or how often a position must be reheded in order to maintain a delta-neutral position.

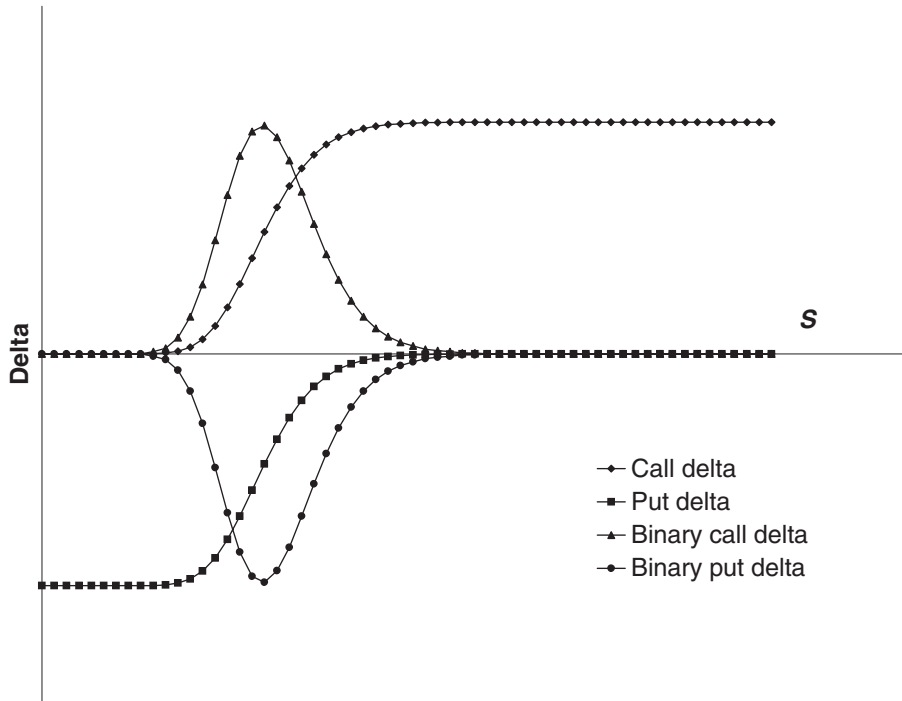


Figure 8.11 The deltas of a call, a put, a binary call and a binary put option. (Binary values scaled to a maximum value of one.)

Although the delta also varies with time this effect is dominated by the Brownian nature of the movement in the underlying.

In a delta-neutral position the gamma is partly responsible for making the return on the portfolio equal to the risk-free rate, the no-arbitrage condition of Chapter 6. The rest of this task falls to the time-derivative of the option value, discussed below. Actually, the situation is far more complicated than this because of the necessary discreteness in the hedging, there is a finite time between rehedges. In any delta-hedged position you make money on some hedges and lose some on others. In a long gamma position ($\Gamma > 0$) you make money on the large moves in the underlying and lose it on the small moves. The net effect is to get the risk-free rate of return on the portfolio.

Gamma also plays an important role when there is a mismatch between the market's view of volatility and the actual volatility of the underlying.

Because costs can be large and because one wants to reduce exposure to model error it is natural to try to minimize the need to rebalance the portfolio too frequently. Since gamma is a measure of sensitivity of the hedge ratio Δ to the movement in the underlying, the hedging requirement can be decreased by a gamma-neutral strategy. This means buying or selling more *options*, not just the underlying. Because the gamma of the underlying (its second derivative) is zero, we cannot add gamma to our position just with the underlying. We can have as many options in our position as we want, we choose the quantities of each such that both delta and gamma are zero. The minimal requirement is

to hold two different types of option and the underlying. In practice, the option position is not readjusted too often because, if the cost of transacting in the underlying is large, then the cost of transacting in its derivatives is even larger.

Here are some formulæ for the gammas of common contracts:



Excel
representation of
gamma values

Gammas of common contracts

Call	$\frac{e^{-D(T-t)}N'(d_1)}{\sigma S\sqrt{T-t}}$
Put	$\frac{e^{-D(T-t)}N'(d_1)}{\sigma S\sqrt{T-t}}$
Binary call	$-\frac{e^{-r(T-t)}d_1N'(d_2)}{\sigma^2 S^2(T-t)}$
Binary put	$\frac{e^{-r(T-t)}d_1N'(d_2)}{\sigma^2 S^2(T-t)}$

Examples of these functions are plotted in Figure 8.12, with some scaling for the binaries.

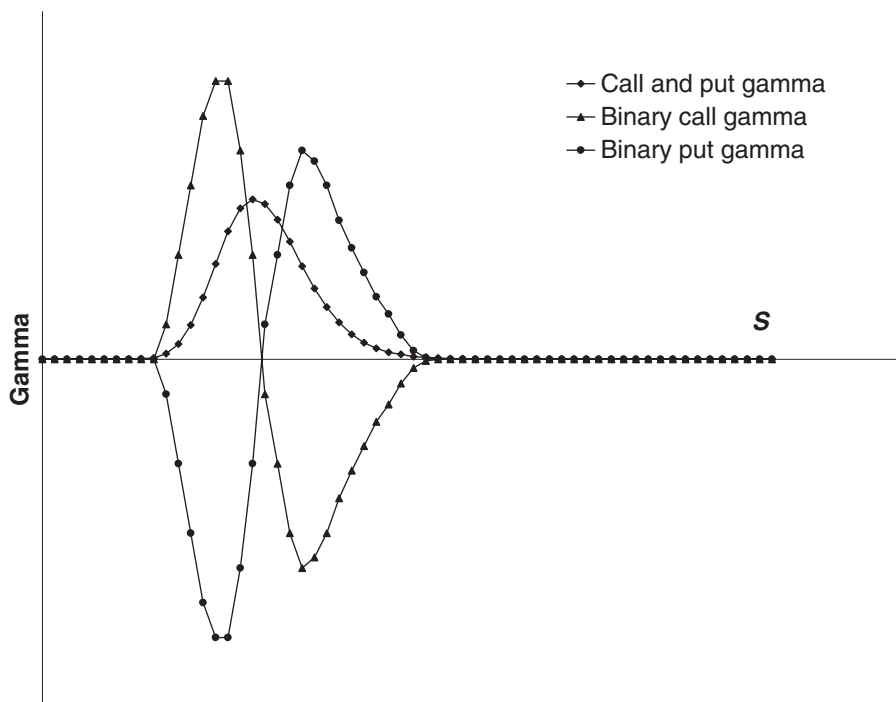


Figure 8.12 The gammas of a call, a put, a binary call and a binary put option.

8.5 THETA

Theta, Θ , is the rate of change of the option price with time.

$$\Theta = \frac{\partial V}{\partial t}$$

The theta is related to the option value, the delta and the gamma by the Black–Scholes equation. In a delta-hedged portfolio the theta contributes to ensuring that the portfolio earns the risk-free rate. But it contributes in a completely certain way, unlike the gamma which contributes the right amount *on average*.

Here are some formulae for the thetas of common contracts:

Thetas of common contracts

$$\text{Call} \quad -\frac{\sigma S e^{-D(T-t)} N'(d_1)}{2\sqrt{T-t}} + D S N(d_1) e^{-D(T-t)} - r E e^{-r(T-t)} N(d_2)$$

$$\text{Put} \quad -\frac{\sigma S e^{-D(T-t)} N'(-d_1)}{2\sqrt{T-t}} - D S N(-d_1) e^{-D(T-t)} + r E e^{-r(T-t)} N(-d_2)$$

$$\text{Binary call} \quad r e^{-r(T-t)} N(d_2) + e^{-r(T-t)} N'(d_2) \left(\frac{d_1}{2(T-t)} - \frac{r-D}{\sigma\sqrt{T-t}} \right)$$

$$\text{Binary put} \quad r e^{-r(T-t)} (1 - N(d_2)) - e^{-r(T-t)} N'(d_2) \left(\frac{d_1}{2(T-t)} - \frac{r-D}{\sigma\sqrt{T-t}} \right)$$



Excel representation of theta values

These functions are plotted in Figure 8.13.

8.6 SPEED

The **speed** of an option is the rate of change of the gamma with respect to the stock price.

$$\text{Speed} = \frac{\partial^3 V}{\partial S^3}$$

Traders use the gamma to estimate how much they will have to rehedge if the stock moves. The stock moves by \$1 so the delta changes by whatever the gamma is. But that's only an approximation. The delta may change by more or less than this, especially if the stock moves by a larger amount, or the option is close to the strike and expiration. Hence the use of speed in a higher-order Taylor series expansion.

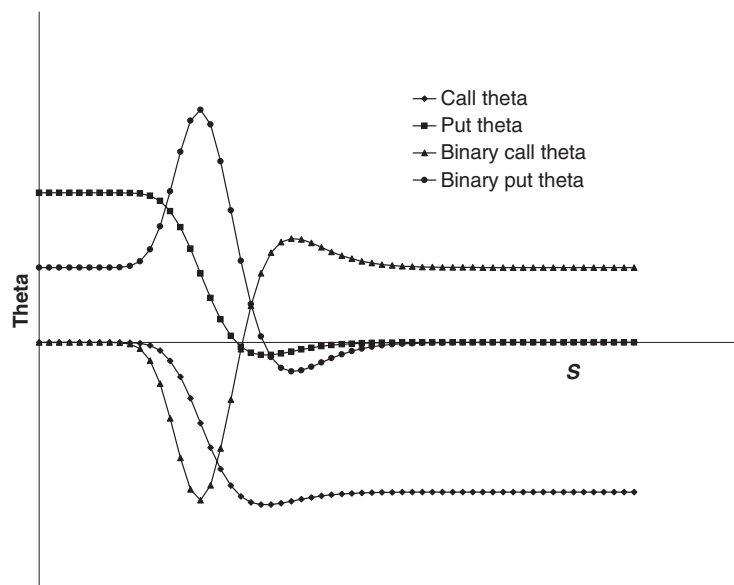
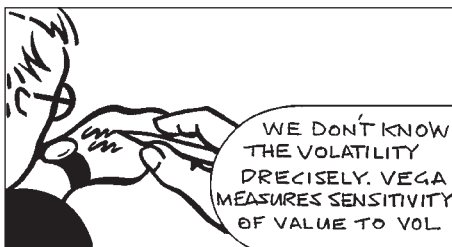


Figure 8.13 The thetas of a call, a put, a binary call and a binary put option.

Here are some formulæ for the speed of common contracts:

Speed of common contracts

Call	$-\frac{e^{-D(T-t)}N'(d_1)}{\sigma^2 S^2(T-t)} \left(d_1 + \sigma\sqrt{T-t} \right)$
Put	$-\frac{e^{-D(T-t)}N'(d_1)}{\sigma^2 S^2(T-t)} \left(d_1 + \sigma\sqrt{T-t} \right)$
Binary call	$-\frac{e^{-r(T-t)}N'(d_2)}{\sigma^2 S^3(T-t)} \left(-2d_1 + \frac{1-d_1d_2}{\sigma\sqrt{T-t}} \right)$
Binary put	$\frac{e^{-r(T-t)}N'(d_2)}{\sigma^2 S^3(T-t)} \left(-2d_1 + \frac{1-d_1d_2}{\sigma\sqrt{T-t}} \right)$



8.7 VEGA

Vega, a.k.a. zeta and kappa, is a very important but confusing quantity. It is the sensitivity of the option price to volatility.

$$\text{Vega} = \frac{\partial V}{\partial \sigma}$$

This is completely different from the other greeks³ since it is a derivative with respect to a parameter and not a variable. This makes something of a difference when we come to find numerical solutions for such quantities.

In practice, the volatility of the underlying is not known with certainty. Not only is it very difficult to measure at any time, it is even harder to predict what it will do in the future. Suppose that we put a volatility of 20% into an option pricing formula, how sensitive is the price to that number? That's the vega.

As with gamma hedging, one can vega hedge to reduce sensitivity to the volatility. This is a major step towards eliminating some model risk, since it reduces dependence on a quantity that, to be honest, is not known very accurately.

There is a downside to the measurement of vega.

It is only really meaningful for options having single-signed gamma everywhere. For example, it makes sense to measure vega for calls and puts but not binary calls and binary puts. I have included the formulae for the vega of such contracts below, but they should be used with care, if at all. The reason for this is that call and put values (and options with single-signed gamma) have values that are monotonic in the volatility: increase the volatility in a call and its value increases everywhere. However,

contracts with a gamma that changes sign may have a vega measured at zero because as we increase the volatility the price may rise somewhere and fall somewhere else. Such a contract is very exposed to volatility risk but that risk is not measured by the vega.

Here are some formulae for the vegas of common contracts:

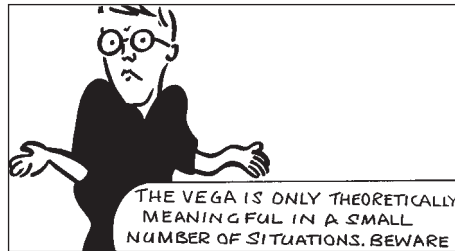
Vegas of common contracts

$$\text{Call} \quad S\sqrt{T-t}e^{-D(T-t)}N'(d_1)$$

$$\text{Put} \quad S\sqrt{T-t}e^{-D(T-t)}N'(d_1)$$

$$\text{Binary call} \quad -e^{-r(T-t)}N'(d_2)\frac{d_1}{\sigma}$$

$$\text{Binary put} \quad e^{-r(T-t)}N'(d_2)\frac{d_1}{\sigma}$$



Excel representation of vega values

In Figure 8.14 is shown the value of an at-the-money call option as a function of the volatility. There is one year to expiry, the strike is 100, the interest rate is 10% and there are no dividends. No matter how far in or out

³ It's not even Greek. Among other things it is an American car, a star (Alpha Lyræ), the real name of Zorro, there are a couple of 16th century Spanish authors called Vega, an op art painting by Vasarely and a character in the computer game 'Street Fighter.' And who could forget Vincent, and his brother, and his 'cousin'?

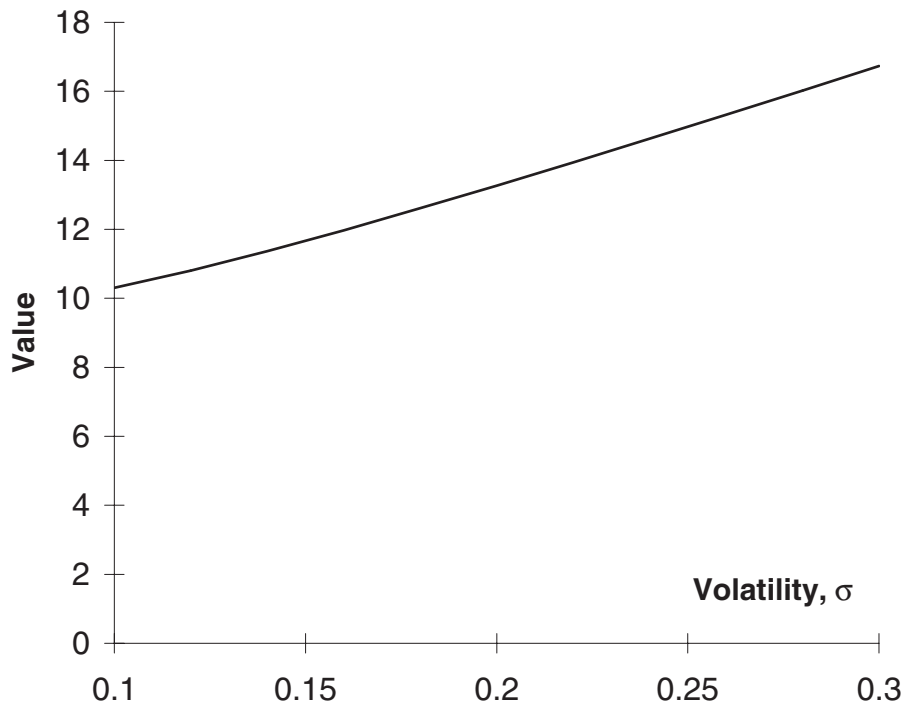


Figure 8.14 The value of an at-the-money call option as a function of volatility.

of the money this curve is always monotonically increasing for call options and put options, uncertainty adds value to the contract. The slope of this curve is the vega.

8.8 RHO

Rho, ρ , is the sensitivity of the option value to the interest rate used in the Black–Scholes formulæ:

$$\rho = \frac{\partial V}{\partial r}$$

In practice one often uses a whole term structure of interest rates, meaning a time-dependent rate $r(t)$. Rho would then be the sensitivity to the level of the rates assuming a parallel shift in rates at all times. Again, you must be careful for which contracts you measure rho.

Here are some formulae for the rhos of common contracts:

Rhos of common contracts

$$\text{Call} \quad E(T-t)e^{-r(T-t)}N(d_2)$$

$$\text{Put} \quad -E(T-t)e^{-r(T-t)}N(-d_2)$$

$$\text{Binary call} \quad -(T-t)e^{-r(T-t)}N(d_2) + \frac{\sqrt{T-t}}{\sigma}e^{-r(T-t)}N'(d_2)$$

$$\text{Binary put} \quad -(T-t)e^{-r(T-t)}(1-N(d_2)) - \frac{\sqrt{T-t}}{\sigma}e^{-r(T-t)}N'(d_2)$$



Excel representation of rho values

The sensitivities of common contract to the dividend yield or foreign interest rate are given by the following formulae:

Sensitivity to dividend for common contracts

$$\text{Call} \quad -(T-t)Se^{-D(T-t)}N(d_1)$$

$$\text{Put} \quad (T-t)Se^{-D(T-t)}N(-d_1)$$

$$\text{Binary call} \quad -\frac{\sqrt{T-t}}{\sigma}e^{-r(T-t)}N'(d_2)$$

$$\text{Binary put} \quad \frac{\sqrt{T-t}}{\sigma}e^{-r(T-t)}N'(d_2)$$

8.9 IMPLIED VOLATILITY

The Black–Scholes formula for a call option takes as input the expiry, the strike, the underlying and the interest rate *together with the volatility* to output the price. All but the volatility are easily measured. How do we know what volatility to put into the formulae? A trader can see on his screen that a certain call option with four months until expiry and a strike of 100 is trading at 6.51 with the underlying at 101.5 and a short-term interest rate of 8%. Can we use this information in some way?

Turn the relationship between volatility and an option price on its head, if we can see the price at which the option is trading, we can ask 'What volatility must I use to get the correct market price?' This is called the **implied volatility**. The implied volatility is the volatility of the underlying which when substituted into the Black–Scholes formula gives a theoretical price equal to the market price. In a sense it is the market's view of volatility over the life of the option. Assuming that we are using call prices to estimate the implied volatility then provided the option price is less than the asset and greater than zero then we can find

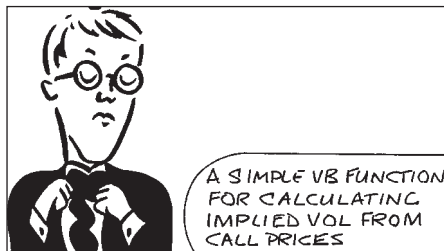
a unique value for the implied volatility. (If the option price is outside these bounds then there's a very extreme arbitrage opportunity.) Because there is no simple formula for the implied volatility as a function of the option value we must solve the equation

$$V_{BS}(S_0, t_0; \sigma, r; E, T) = \text{known value}$$

for σ , where V_{BS} is the Black-Scholes formula. Today's asset price is S_0 , the date is t_0 and everything is known in this equation except for σ . Below is an algorithm for finding the implied volatility from the market price of a call option to any required degree of accuracy. The method used is **Newton-Raphson** which uses the derivative of the option price with respect to the volatility (the vega) in the calculation. This method is particularly good for such a well-behaved function as a call value.



VB implementation



```
Function ImpVolCall(MktPrice As Double, _
    Strike As Double, Expiry As _
    Double, Asset As Double, _
    IntRate As Double, error _
    As Double)
    Volatility = 0.2
    dv = error + 1
    While Abs(dv) > error
        d1 = Log(Asset / Strike) + (IntRate _
            + 0.5 * Volatility * Volatility) _
            * Expiry
        d1 = d1 / (Volatility * Sqr(Expiry))
        d2 = d1 - Volatility * Sqr(Expiry)
        PriceError = Asset * cdf(d1) - Strike _
            * Exp(-IntRate * Expiry) _
            * cdf(d2) - MktPrice
        Vega = Asset * Sqr(Expiry / 3.1415926 _
            / 2) * Exp(-0.5 * d1 * d1)
        dv = PriceError / Vega
        Volatility = Volatility - dv
    Wend
    ImpVolCall = Volatility
End Function
```

In this we need the cumulative distribution function for the Normal distribution. The following is a simple algorithm which gives an accurate, and fast, approximation to the cumulative distribution function of the standardized Normal:

$$\text{For } x \geq 0 \quad N(x) \approx 1 - \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \left(a_1 d + a_2 d^2 + a_3 d^3 + a_4 d^4 + a_5 d^5 \right)$$

where

$$d = \frac{1}{1 + 0.02316419x}$$

and

$$a_1 = 0.31938153, \quad a_2 = -0.356563782, \quad a_3 = 1.781477937, \\ a_4 = -1.821255978 \quad \text{and} \quad a_5 = 1.330274429.$$

For $x < 0$ use the fact that $N(x) + N(-x) = 1$.

```

Function cdf(x As Double) As Double
Dim d As Double
Dim temp as Double
Dim a1 As Double
Dim a2 As Double
Dim a3 As Double
Dim a4 As Double
Dim a5 As Double
d = 1 / (1 + 0.2316419 * Abs(x))
a1 = 0.31938153
a2 = -0.356563782
a3 = 1.781477937
a4 = -1.821255978
a5 = 1.330274429
temp = a5
temp = a4 + d * temp
temp = a3 + d * temp
temp = a2 + d * temp
temp = a1 + d * temp
temp = d * temp
cdf = 1 - 1 / Sqr(2 * 3.1415926) * Exp(-0.5 * x * x) * temp
If x < 0 Then cdf = 1 - cdf
End Function

```



VB implementation

In practice if we calculate the implied volatility for many different strikes and expiries on the same underlying then we find that *the volatility is not constant*. A typical result is that of Figure 8.15 which shows the implied volatilities for the S&P500 on 9th September

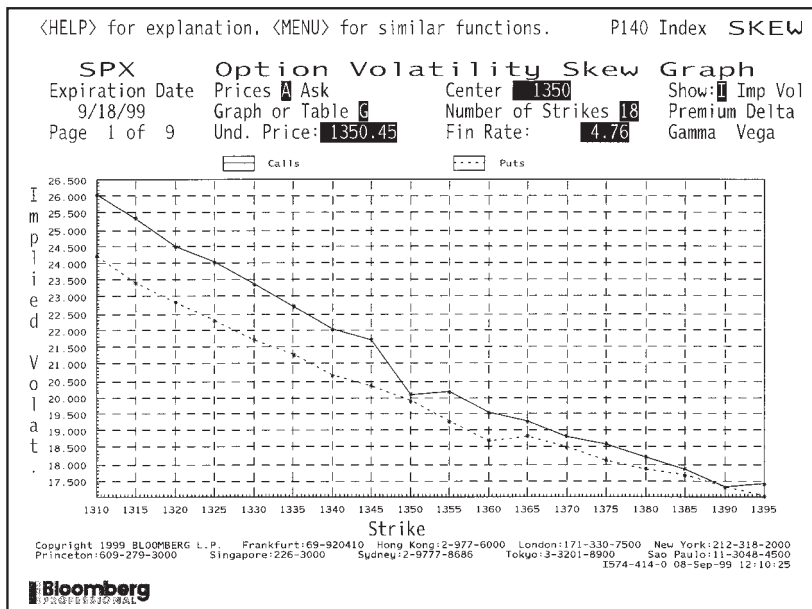
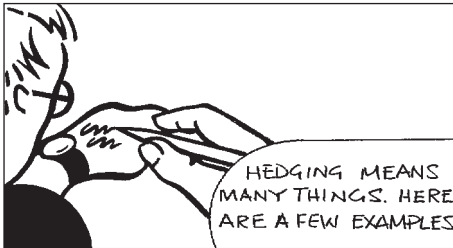


Figure 8.15 Implied volatilities for the S&P500. Source: Bloomberg L.P.

1999 for options expiring later in the month. The implied volatilities for the calls and puts should be identical, because of put-call parity. The differences seen here could be due to bid-offer spread or calculations performed at slightly different times.

This shape is commonly referred to as the **smile**, but it could also be in the shape of a **frown**. In this example it's a rather lopsided wry grin. Whatever the shape, it tends to persist with time, with certain shapes being characteristic of certain markets.

The dependence of the implied volatility on strike and expiry can be interpreted in many ways. The easiest interpretation is that it represents the market's view of future volatility in some complex way.



8.10 **A CLASSIFICATION OF HEDGING TYPES**

8.10.1 Why hedge?

'Hedging' in its broadest sense means the reduction of risk by exploiting relationships or correlation between various risky investments (or bets). The concept is used widely in horse racing, other sports betting and, of course, high finance. The reason for hedging is that it can lead to an improved risk/return. In the classical Modern Portfolio Theory framework (Chapter 21), for example, it is usually possible to construct many portfolios having the same expected return but with different variance of returns ('risk'). Clearly, if you have two portfolios with the same expected return the one with the lower risk is the better investment.

8.10.2 The two main classifications

Probably the most important distinction between types of hedging is between model-independent and model-dependent hedging strategies.

Model-independent hedging: An example of such hedging is put-call parity. There is a simple relationship between calls and puts on an asset (when they are both European and with the same strikes and expiries), the underlying stock and a zero-coupon bond with the same maturity. This relationship is completely independent of how the underlying asset changes in value. Another example is spot-forward parity. In neither case do we have to specify the dynamics of the asset, not even its volatility, to find a possible hedge. Such model-independent hedges are few and far between.

Model-dependent hedging: Most sophisticated finance hedging strategies depend on a model for the underlying asset. The obvious example is the hedging used in the Black–Scholes analysis that leads to a whole theory for the value of derivatives. In pricing derivatives we typically need to at least know the volatility of the underlying asset. If the model is wrong then the option value and any hedging strategy will also be wrong.

8.10.3 Delta hedging

One of the building blocks of derivatives theory is **delta hedging**. This is the theoretically perfect elimination of all risk by using a very clever hedge between the option and its underlying. Delta hedging exploits the perfect correlation between the changes in the option value and the changes in the stock price. This is an example of 'dynamic' hedging; the hedge must be continually monitored and frequently adjusted by the sale or purchase of the underlying asset. Because of the frequent rehedge, any dynamic hedging strategy is going to result in losses due to transaction costs. In some markets this can be very important.

8.10.4 Gamma hedging

To reduce the size of each rehedged and/or to increase the time between rehedges, and thus reduce costs, the technique of **gamma hedging** is often employed. A portfolio that is delta hedged is insensitive to movements in the underlying as long as those movements are quite small. There is a small error in this due to the convexity of the portfolio with respect to the underlying. Gamma hedging is a more accurate form of hedging that theoretically eliminates these second-order effects. Typically, one hedges one, exotic, say, contract with a vanilla contract and the underlying. The quantities of the vanilla and the underlying are chosen so as to make both the portfolio delta and the portfolio gamma instantaneously zero.

8.10.5 Vega hedging

As I said above, the prices and hedging strategies are only as good as the model for the underlying. The key parameter that determines the value of a contract is the volatility of the underlying asset. Unfortunately, this is a very difficult parameter to measure or even estimate. Nor is it usually a constant as assumed in the simple theories. Obviously, the value of a contract depends on this parameter, and so to ensure that our portfolio value is insensitive to this parameter we can **vega hedge**. This means that we hedge one option with both the underlying and another option in such a way that both the delta and the vega, the sensitivity of the portfolio value to volatility, are zero. This is often quite satisfactory in practice but is usually theoretically inconsistent; we should not use a constant volatility (basic Black–Scholes) model to calculate sensitivities to parameters that are assumed not to vary. The distinction between variables (underlying asset price and time) and parameters (volatility, dividend yield, interest rate) is extremely important here. It is justifiable to rely on sensitivities of prices to variables, but usually not sensitivity to parameters. To get around this problem it is possible to independently model volatility, etc. as variables themselves. In such a way it is possible to build up a consistent theory.

8.10.6 Static hedging

There are quite a few problems with delta hedging, on both the practical and the theoretical side. In practice, hedging must be done at discrete times and is costly. Sometimes

one has to buy or sell a prohibitively large number of the underlying in order to follow the theory. This is a problem with barrier options and options with discontinuous payoff. On the theoretical side, the model for the underlying is not perfect, at the very least we do not know parameter values accurately. Delta hedging alone leaves us very exposed to the model; this is model risk. Many of these problems can be reduced or eliminated if we follow a strategy of **static hedging** as well as delta hedging: buy or sell more liquid traded contracts to reduce the cashflows in the original contract. The static hedge is put into place now, and left until expiry. In the extreme case where an exotic contract has all of its cashflows matched by cashflows from traded options then its value is given by the cost of setting up the static hedge; a model is not needed.

8.10.7 Margin hedging

Often what causes banks, and other institutions, to suffer during volatile markets is not the change in the paper value of their assets but the requirement to suddenly come up with a large amount of cash to cover an unexpected margin call. Recent examples where margin has caused significant damage are Metallgesellschaft and Long Term Capital Management. Writing options is very risky. The downside of buying an option is just the initial premium, the upside may be unlimited. The upside of writing an option is limited, but the downside could be huge. For this reason, to cover the risk of default in the event of an unfavorable outcome, the clearing houses that register and settle options insist on the deposit of a margin by the writers of options. Margin comes in two forms, the initial margin and the maintenance margin. The initial margin is the amount deposited at the initiation of the contract. The total amount held as margin must stay above a prescribed maintenance margin. If it ever falls below this level then more money (or equivalent in bonds, stocks, etc.) must be deposited. The amount of margin that must be deposited depends on the particular contract. A dramatic market move could result in a sudden large margin call that may be difficult to meet. To prevent this situation it is possible to **margin hedge**. That is, set up a portfolio such that margin calls on one part of the portfolio are balanced by refunds from other parts. Usually over-the-counter contracts have no associated margin requirements and so won't appear in the calculation.

8.10.8 Crash (Platinum) hedging

The final variety of hedging that we discuss is specific to extreme markets. Market crashes have at least two obvious effects on our hedging. First of all, the moves are so large and rapid that they cannot be traditionally delta hedged. The convexity effect is not small. Second, normal market correlations become meaningless. Typically all correlations become one (or minus one). **Crash** or **Platinum hedging** exploits the latter effect in such a way as to minimize the worst possible outcome for the portfolio. The method, called CrashMetrics (Chapter 25), does not rely on difficult to measure parameters such as volatilities and so is a very robust hedge. Platinum hedging comes in two types: hedging the paper value of the portfolio and hedging the margin calls.

8.11 SUMMARY

In this chapter we went through the derivation of some of the most important formulæ. We also saw the definitions and descriptions of the hedge ratios. Trading in derivatives

would be no more than gambling if you took away the ability to hedge. Hedging is all about managing risk and reducing uncertainty.

FURTHER READING

- See Taleb (1997) for a lot of detailed analysis of vega.
- See Press *et al.* (1992) for more routines for finding roots, i.e. for finding implied volatilities.
- There are many 'virtual' option pricers on the internet. See, for example, www.cboe.com.
- I'm not going to spend much time on deriving or even presenting formulae. There are 1001 books that contain option formulae, there is even one book with 1001 formulae (Haug, 2007).
- See the series of articles by Thorp (2002) on how he derived the correct option pricing formulae in the late 1960s.
- Haug (2003) discusses the sophisticated trader use of the simple equations.
- For the definitions of even more greeks see Wilmott (2006).

Time Out...

Another look at Black–Scholes

Let's take another look at the Black–Scholes equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0.$$

The option value $V(S, t)$ depends on (or 'is a function of') the asset price S and the time t .

The first derivative of the option value with respect to time is called the theta:

$$\Theta = \frac{\partial V}{\partial t}.$$

Notice that this is a partial derivative and so theta is the gradient of the option value in the direction of changing time, asset price fixed. It measures the rate of change of the option value with time if the asset price doesn't move, hence the other name 'time decay.'

The first derivative of the option value with respect to the asset price is called the delta:

$$\Delta = \frac{\partial V}{\partial S}.$$



This is the slope in the S direction with time fixed. Asset prices change very rapidly and so the dominant change in the option value from moment to moment is the delta multiplied by the change in the asset price. This is just a simple application of Taylor series; the difference between the option price at time t when the asset is at S and a later time $t + \delta t$ when the asset price is $S + \delta S$ is given by

$$V(S + \delta S, t + \delta t) - V(S, t) = \Delta \delta S + \dots$$

The $\dots$ are terms which are, generally speaking, smaller than the leading term. They are still important, as we'll see in a moment.

Because the change in option value and the change in asset price are so closely linked we can see that holding a quantity Δ of the underlying asset short we can eliminate, to leading order, fluctuations in our net portfolio value. This is the basis of delta hedging.

The second derivative of the option value with respect to the asset price is called the gamma:

$$\Gamma = \frac{\partial^2 V}{\partial S^2}.$$

This is also just the S derivative of the delta. If the asset changes by an amount δS then the delta changes by an amount $\Gamma \delta S$. Thus the gamma is a measure of how much one might have to re hedge, and gives a measure of the amount of transaction costs from delta hedging.

Now we can interpret all the terms in the Black–Scholes equation, but what does the equation itself mean?

Written in terms of the greeks, the Black–Scholes equation is

$$\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta - rV = 0.$$

Reordering this we have

$$\Theta = rV - rS\Delta - \frac{1}{2}\sigma^2 S^2 \Gamma = r(V - S\Delta) - \frac{1}{2}\sigma^2 S^2 \Gamma.$$

When we have a delta-hedged position we hold the option with value V and are short Δ of the underlying asset. Thus our portfolio value is at any time

$$V - S\Delta.$$

So we can write the Black–Scholes equation in words as

$$\begin{aligned} \text{Time decay} &= \text{interest received on cash equivalent of portfolio value} \\ &\quad - \frac{1}{2}\sigma^2 S^2 \text{gamma.} \end{aligned}$$

The option value grows by an equivalent of interest that would have been received by a riskless pure cash position. But the delta-hedged option is not a cash position. That's where the final, gamma, term comes in.

Ignoring the interest on the cash equivalent, the theta and gamma terms add up to zero. Of course, you can't ignore this interest unless the portfolio has zero value or rates are zero.

The delta hedge is only accurate to leading order. If one is hedging with finite time intervals between rehedges then there is inevitably a little bit of randomness that we can't hedge away. We can see this if we go to higher order in the Taylor series expansion of $V(S + \delta S, t + \delta t)$:

$$V(S + \delta S, t + \delta t) - V(S, t) = \Delta \delta S + \Theta \delta t + \frac{1}{2} \Gamma \delta S^2 \dots$$

The Θ term is predictable if we know the time δt between hedges (and it has already appeared in the Black–Scholes equation). But the Γ term is multiplied by the random δS^2 . We can't hedge this away perfectly. It is, in practice, the source of hedging errors. However, if we re hedge sufficiently frequently (i.e. δt is very small) then the combined effect of the gamma terms is via an *average* of the δS^2 . And this average is $\sigma^2 S^2 \delta t$. Why is it the average that matters? It's like betting on the toss of a biased coin. If you have an advantage then you can exploit it by betting a small amount but very, very often. In the long run you will certainly win. (In terms of standard deviations, as the time between hedges decreases so does the standard deviation of the hedging error accumulated over the life of the option.)

We can now see that the gamma term in the Black–Scholes equation is to balance the higher-order fluctuations in the option value. Naturally, it therefore depends on the magnitude of these fluctuations, the volatility of the underlying asset.

	Call	Put	Binary Call	Binary Put
Value V Black-Scholes value	$Se^{-D(T-t)}N(d_1) - Ee^{-r(T-t)}N(d_2)$	$-Se^{-D(T-t)}N(-d_1) + Ee^{-r(T-t)}N(-d_2)$	$e^{-r(T-t)}N(d_2)$	$e^{-r(T-t)}(1 - N(d_2))$
Delta $\frac{\partial V}{\partial S}$ Sensitivity to underlying	$e^{-D(T-t)}N(d_1)$	$e^{-D(T-t)}(N(d_1) - 1)$	$\frac{e^{-r(T-t)}N(d_2)}{\sigma\sqrt{T-t}}$	$-\frac{e^{-r(T-t)}N(d_2)}{\sigma\sqrt{T-t}}$
Gamma $\frac{\partial^2 V}{\partial S^2}$ Sensitivity of delta to underlying	$\frac{e^{-D(T-t)}N(d_1)}{\sigma\sqrt{T-t}}$	$\frac{e^{-D(T-t)}N(d_1)}{\sigma\sqrt{T-t}}$	$-\frac{e^{-r(T-t)}d_1N'(d_2)}{\sigma^2S^2(T-t)}$	$\frac{e^{-r(T-t)}d_1N'(d_2)}{\sigma^2S^2(T-t)}$
Theta $\frac{\partial V}{\partial t}$ Sensitivity to time	$-\frac{\sigma Se^{-D(T-t)}N'(d_1)}{2\sqrt{T-t}} + DSN(d_1)e^{-D(T-t)} - rEe^{-r(T-t)}N(d_2)$	$-\frac{\sigma Se^{-D(T-t)}N'(-d_1)}{2\sqrt{T-t}} - DSN(-d_1)e^{-D(T-t)} + rEe^{-r(T-t)}N(-d_2)$	$re^{-r(T-t)}N(d_2) + e^{-r(T-t)}N(d_2) \times \left(\frac{d_1}{2(T-t)} - \frac{r-D}{\sigma\sqrt{T-t}} \right)$	$re^{-r(T-t)}(1 - N(d_2)) - e^{-r(T-t)}N(d_2) \times \left(\frac{d_1}{2(T-t)} - \frac{r-D}{\sigma\sqrt{T-t}} \right)$
Speed $\frac{\partial^3 V}{\partial S^3}$ Sensitivity of gamma to underlying	$-\frac{e^{-D(T-t)}N'(d_1)}{\sigma^2S^2(T-t)} \times (d_1 + \sigma\sqrt{T-t})$	$-\frac{e^{-D(T-t)}N'(d_1)}{\sigma^2S^2(T-t)} \times (d_1 + \sigma\sqrt{T-t})$	$-\frac{e^{-r(T-t)}N'(d_2)}{\sigma^2S^3(T-t)} \times \left(-2d_1 + \frac{1-d_1d_2}{\sigma\sqrt{T-t}} \right)$	$\frac{e^{-r(T-t)}N'(d_2)}{\sigma^2S^3(T-t)} \times \left(-2d_1 + \frac{1-d_1d_2}{\sigma\sqrt{T-t}} \right)$
Vega $\frac{\partial V}{\partial \sigma}$ Sensitivity to volatility	$S\sqrt{T-t}e^{-D(T-t)}N'(d_1)$	$S\sqrt{T-t}e^{-D(T-t)}N'(d_1)$	$-\frac{e^{-r(T-t)}N'(d_2)}{\sigma} \times \left(\sqrt{T-t} + \frac{d_2}{\sigma} \right)$	$\frac{e^{-r(T-t)}N'(d_2)}{\sigma} \times \left(\sqrt{T-t} + \frac{d_2}{\sigma} \right)$
Rho (r) $\frac{\partial V}{\partial r}$ Sensitivity to interest rate	$E(T-t)e^{-r(T-t)}N(d_2)$	$-E(T-t)e^{-r(T-t)}N(-d_2)$	$-\frac{(T-t)e^{-r(T-t)}N(d_2)}{\sigma} + \frac{\sqrt{T-t}}{\sigma}e^{-r(T-t)}N'(d_2)$	$-\frac{(T-t)e^{-r(T-t)}(1 - N(d_2))}{\sigma} - \frac{\sqrt{T-t}}{\sigma}e^{-r(T-t)}N'(d_2)$
Rho (D) $\frac{\partial V}{\partial D}$ Sensitivity to dividend yield	$-(T-t)Se^{-D(T-t)}N(d_1)$	$(T-t)Se^{-D(T-t)}N(-d_1)$	$-\frac{\sqrt{T-t}}{\sigma}e^{-r(T-t)}N'(d_2)$	$\frac{\sqrt{T-t}}{\sigma}e^{-r(T-t)}N'(d_2)$

$$d_1 = \frac{\log\left(\frac{S}{E}\right) + (r - D + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = \frac{\log\left(\frac{S}{E}\right) + (r - D - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} = d_1 - \sigma\sqrt{T-t}, \quad N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}\xi^2} d\xi \quad \text{and} \quad N'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

EXERCISES

- Find the explicit solution for the value of a European option with payoff $\Lambda(S)$ and expiry at time T , where

$$\Lambda(S) = \begin{cases} S & \text{if } S > E \\ 0 & \text{if } S < E. \end{cases}$$

- Find the explicit solution for the value of a European supershare option, with expiry at time T and payoff

$$\Lambda(S) = \mathcal{H}(S - E_1) - \mathcal{H}(S - E_2),$$

where $E_1 < E_2$.

- Consider the pay-later call option. This has payoff $\Lambda(S) = \max(S - E, 0)$ at time T . The holder of the option does not pay a premium when the contract is set up, but must pay Q to the writer at expiry, only if $S \geq E$. What is the value of Q ?
- Find the implied volatility of the following European call. The call has four months until expiry and an exercise price of \$100. The call is worth \$6.51 and the underlying trades at \$101.5, discount using a short-term risk-free continuously compounding interest rate of 8% per annum.
- Consider a European call, currently at the money. Why is delta hedging self-financing in the following situations?
 - The share price rises until expiry,
 - The share price falls until expiry.
- Using the explicit solutions for the European call and put options, check that put-call parity holds.
- Consider an asset with zero volatility. We can explicitly calculate the future value of the asset and hence that of a call option with the asset as the underlying. The value of the call option will then depend on the growth rate of the asset, μ . On the other hand, we can use the explicit formula for the call option, in which μ does not appear. Explain this apparent contradiction.
- The range forward contract is specified as follows: at expiry, the holder must buy the asset for E_1 if $S < E_1$, for S if $E_1 \leq S \leq E_2$ and for E_2 if $S > E_2$. Find the relationship between E_1 and E_2 when the initial value of the contract is zero and $E_1 < E_2$.
- A forward start call option is specified as follows: at time T_1 , the holder is given a European call option with exercise price $S(T_1)$ and expiry at time $T_1 + T_2$. What is the value of the option for $0 \leq t \leq T_1$?
- Consider a delta-neutral portfolio of derivatives, Π . For a small change in the price of the underlying asset, δS , over a short time interval, δt , show that the change in the portfolio value, $\delta \Pi$, satisfies

$$\delta \Pi = \Theta \delta t + \frac{1}{2} \Gamma \delta S^2,$$

where $\Theta = \frac{\partial \Pi}{\partial t}$ and $\Gamma = \frac{\partial^2 \Pi}{\partial S^2}$.

11. Show that for a delta-neutral portfolio of options on a non-dividend paying stock, Π ,

$$\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = r\Pi.$$

12. Show that the vega of an option, v , satisfies the differential equation

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 v}{\partial S^2} + rS \frac{\partial v}{\partial S} - rv + \sigma S^2 \Gamma = 0,$$

where $\Gamma = \frac{\partial^2 V}{\partial S^2}$. What is the final condition?

13. Find the partial differential equation satisfied by ρ , the sensitivity of the option value to the interest rate.
14. Use put-call parity to find the relationships between the deltas, gammas, vegas, thetas and rhos of European call and put options.
15. The fundamental solution, u_δ , is the solution of the diffusion equation on $-\infty < x < \infty$ and $\tau > 0$ with $u(x, 0) = \delta(x)$. Use this solution to solve the more general problem:

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}, \text{ on } -\infty < x < \infty, \tau > 0,$$

with $u(x, 0) = u_0(x)$.